METABOLIC & FAT LOSS

AOD-9604 — the full review

GROWTH HORMONE FRAGMENT (176-191)

This is the deeper look behind the short version — what the pivotal human trial found, why the selectivity claim holds up and the fat-loss claim does not, the bone and liver signals in the animal data, and what your consultation covers.

What it is

AOD-9604 is a laboratory-made piece of human growth hormone — specifically the fifteen-amino-acid tail of the molecule, residues 176-191. Growth hormone burns fat but also raises blood sugar and drives tissue growth elsewhere in the body; AOD-9604 was designed to isolate the fat-burning end of the growth hormone molecule and leave the rest behind.

It is sold for fat loss and, more recently, for joint support, sometimes marketed under the name HGH Fragment 176-191.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Fat loss and weight loss	**Human trials**	Taken through a full 24-week randomised trial in several hundred people — the result is in the full review.
Fat loss without growth hormone's other effects	**Animal data only**	Animal work supports the selectivity: it does not raise blood sugar or IGF-1 the way growth hormone does.
Joint and cartilage support	**Animal data only**	Comes from a rabbit osteoarthritis study; no human joint study exists.

How it works

Growth hormone acts on fat tissue partly through a region at the tail end of the molecule, separate from the region responsible for its growth-promoting and blood-sugar-raising effects elsewhere in the body. AOD-9604 reproduces that tail-end fragment on its own, with the aim of retaining the fat-metabolizing activity while leaving out the parts of growth hormone associated with elevated IGF-1 and blood sugar.

In animal studies, this selectivity has held up: AOD-9604 has not been shown to raise blood sugar or IGF-1 the way intact growth hormone does, which is the basis for it being marketed as a lower-risk fat-loss alternative to growth hormone itself.

What the research actually looked at

AOD-9604 went through a full 24-week randomized, placebo-controlled human trial in several hundred people — the largest and most rigorous human study behind any compound in this category. The trial did not find a statistically significant difference in weight loss between AOD-9604 and placebo, and development of the compound as a weight-loss drug was discontinued after that result.

Separately, a rabbit study looked at cartilage and joint outcomes, which is the basis for the more recent joint-support marketing — no human joint trial has been conducted. The animal safety data on selectivity (not raising IGF-1 or blood sugar) is considered the most solid part of the record.

Common misunderstandings

- **“It's a proven fat-loss treatment because it went through human trials.”** It went through the largest human trial of any compound in this category, and that trial found no significant weight-loss benefit over placebo.
- **“It's basically HGH, just safer.”** It is a small fragment of growth hormone, not growth hormone itself, and reproduces only part of the parent molecule's activity.
- **“The joint-support use has been studied in people.”** The joint and cartilage evidence comes from a single rabbit osteoarthritis study; no human joint trial exists.
- **“Since the fat-loss trial failed, the safety data doesn't matter.”** The trial's safety findings — no significant rise in blood sugar or IGF-1 — are considered a genuine, separate result from its lack of efficacy.

Who it is not for

- Anyone with liver disease or abnormal liver tests.
- Anyone with osteoporosis, osteopenia or a fracture history.
- Anyone being offered it as a cream or a capsule — neither route has human exposure data.
- Anyone with an immune or allergic history, if an injectable is proposed.

Competitive athletes: AOD-9604 is derived from human growth hormone and falls within WADA's S2 category of prohibited peptide hormones, growth factors, and related substances; it is prohibited at all times in tested sport.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Never approved, and the largest trial failed. Tested in several hundred people, no significant weight loss versus placebo, development stopped.

The safety record is actually the reassuring part. Pooled trial data suggests it does not raise IGF-1 or disturb blood sugar the way growth hormone does. That selectivity claim holds up. What does not hold up is the claim that it produces meaningful fat loss.

Reasonably well tolerated, reasonably well studied, and the studies did not find the effect it is sold for.

Clinical detail

Reproduces amino acids 176-191, the C-terminal fragment of human growth hormone.

Tested in a 24-week randomized, placebo-controlled trial in several hundred participants for weight loss; no significant benefit found and development stopped.

Animal data indicates it does not raise IGF-1 or blood glucose the way intact growth hormone does.

Joint and cartilage evidence comes from a single rabbit osteoarthritis study; no human joint trial exists.

Also marketed as HGH Fragment 176-191.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

AOD-9604 is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.